

Give The Top 100 Questions form python which most probability as in interview basic to advance

Since you're preparing for **software development, backend, AI/Data Science, and placements**, here are the **Top 100 Python Interview Questions** arranged from **Basic → Intermediate → Advanced**.

These cover the questions most frequently asked in interviews at companies ranging from startups to major tech firms.

Level 1: Python Basics (1–25)

1. What is Python?
2. What are the features of Python?
3. Why is Python called an interpreted language?
4. Difference between compiled and interpreted languages.
5. What are Python keywords?
6. What are identifiers?
7. What are variables in Python?
8. What are Python data types?
9. Difference between List, Tuple, Set, and Dictionary.
10. Difference between Mutable and Immutable objects.
11. What is type casting?
12. Explain input() and print().
13. Difference between == and is.
14. Difference between = and ==.
15. What is None?
16. What are comments?
17. What is indentation?
18. What are escape characters?
19. What are operators in Python?
20. Difference between / and //.
21. What is ** operator?
22. Explain membership operators.
23. Explain identity operators.
24. What is operator precedence?
25. What are Boolean values?

Level 2: Strings (26–35)

- 26. What are strings?**
- 27. Are strings mutable?**
- 28. String slicing.**
- 29. String indexing.**
- 30. Reverse a string.**
- 31. Check palindrome.**
- 32. Count vowels.**
- 33. Most used string methods.**
- 34. Difference between split() and join().**
- 35. Difference between strip(), lstrip(), and rstrip().**

Level 3: Lists, Tuples, Sets & Dictionaries (36–50)

- 36. List vs Tuple.**
- 37. List methods.**
- 38. Tuple methods.**
- 39. Set methods.**
- 40. Dictionary methods.**
- 41. Difference between append() and extend().**
- 42. Difference between remove(), pop(), del().**
- 43. Difference between sorted() and sort().**
- 44. Merge two dictionaries.**
- 45. Remove duplicates from a list.**
- 46. Find the second largest number.**
- 47. Find frequency of elements.**
- 48. Convert list to tuple.**
- 49. Dictionary comprehension.**
- 50. List comprehension.**

Level 4: Functions (51–60)

- 51. What is a function?
 - 52. Types of functions.
 - 53. Parameters vs Arguments.
 - 54. Default arguments.
 - 55. Keyword arguments.
 - 56. Variable-length arguments (*args, **kwargs).
 - 57. Lambda function.
 - 58. map(), filter(), reduce().
 - 59. Recursive function.
 - 60. Scope of variables (LEGB Rule).
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Level 5: Object-Oriented Programming (61–75)

- 61. What is OOP?
 - 62. Class and Object.
 - 63. Constructor (init).
 - 64. self keyword.
 - 65. Instance vs Class variables.
 - 66. Types of methods.
 - 67. Inheritance.
 - 68. Types of inheritance.
 - 69. Polymorphism.
 - 70. Method Overloading.
 - 71. Method Overriding.
 - 72. Encapsulation.
 - 73. Abstraction.
 - 74. super() keyword.
 - 75. Magic (Dunder) methods.
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Level 6: Exception Handling (76–82)

- 76. What is Exception Handling?
- 77. try-except.

- 78. else and finally.
 - 79. raise keyword.
 - 80. Custom exceptions.
 - 81. Multiple exceptions.
 - 82. Difference between Error and Exception.
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Level 7: File Handling (83–88)

- 83. File modes.
 - 84. read(), readline(), readlines().
 - 85. write() and writelines().
 - 86. with statement.
 - 87. CSV file handling.
 - 88. JSON file handling.
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Level 8: Advanced Python (89–100)

- 89. What are modules?
 - 90. What are packages?
 - 91. Difference between module and package.
 - 92. Decorators.
 - 93. Generators.
 - 94. Iterators.
 - 95. Yield vs Return.
 - 96. Shallow Copy vs Deep Copy.
 - 97. Multithreading vs Multiprocessing.
 - 98. Global Interpreter Lock (GIL).
 - 99. Virtual Environment.
 - 100. Python Memory Management & Garbage Collection.
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Most Frequently Asked Coding Questions

These are the coding questions that appear very often in interviews:

1. Reverse a String

2. Check Palindrome
3. Fibonacci Series
4. Prime Number
5. Armstrong Number
6. Factorial
7. Swap Two Numbers
8. Find Maximum Number
9. Find Second Largest
10. Remove Duplicates
11. Count Vowels
12. Count Digits
13. Find Missing Number
14. Merge Two Lists
15. Find Common Elements
16. Find Frequency of Characters
17. Sort a List
18. Binary Search
19. Linear Search
20. Bubble Sort
21. Selection Sort
22. Insertion Sort
23. Merge Sort
24. Quick Sort
25. Matrix Multiplication

Advanced Coding Questions

26. Anagram Check
27. Longest Common Prefix
28. Longest Substring Without Repeating Characters
29. Valid Parentheses
30. Two Sum

31. Three Sum
 32. Rotate Array
 33. Rotate Matrix
 34. Merge Intervals
 35. Sliding Window Maximum
 36. Maximum Subarray (Kadane's Algorithm)
 37. Product of Array Except Self
 38. Trapping Rain Water
 39. Container With Most Water
 40. LRU Cache
 41. Linked List Reversal
 42. Detect Cycle in Linked List
 43. Merge Two Sorted Linked Lists
 44. Binary Tree Traversals
 45. Lowest Common Ancestor
 46. BFS
 47. DFS
 48. Dijkstra's Algorithm
 49. Topological Sort
 50. Dynamic Programming (Climbing Stairs, Coin Change, Knapsack)
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Must-Know Built-in Functions

- `len()`
- `type()`
- `id()`
- `range()`
- `enumerate()`
- `zip()`
- `sorted()`
- `reversed()`
- `sum()`

- `max()`
- `min()`
- `abs()`
- `all()`
- `any()`
- `map()`
- `filter()`
- `reduce()`
- `eval()`
- `exec()`
- `isinstance()`

Interview Preparation Tip

For placement interviews, aim to master these areas in order:

1. **Python Basics** (variables, data types, operators)
2. **Strings, Lists, Tuples, Sets, Dictionaries**
3. **Functions and Recursion**
4. **OOP**
5. **Exception Handling and File Handling**
6. **Modules, Packages, Iterators, Generators, Decorators**
7. **Multithreading, Multiprocessing, GIL, Memory Management**
8. **50+ Coding Problems** covering arrays, strings, linked lists, trees, graphs, and dynamic programming.

If you can confidently answer these **100 theory questions** and solve the **50 coding problems**, you'll be well prepared for Python interviews ranging from entry-level roles to experienced developer positions.

now i want to be quading and theory top 50 question for each topic of fo;;owing Python Basics (variables, data types, operators) Strings, Lists, Tuples, Sets, Dictionaries Functions and Recursion OOP Exception Handling and File Handling Modules, Packages, Iterators, Generators, Decorators Multithreading, Multiprocessing, GIL, Memory Management

That's an excellent approach. Based on your goal (placements + backend + AI + software development), here's what I'd recommend.

If I generate **50 theory + 50 coding questions for each topic**, the total becomes:

Topic	Theory Coding Total		
Python Basics	50	50	100
Strings, Lists, Tuples, Sets, Dictionaries	50	50	100
Functions & Recursion	50	50	100
OOP	50	50	100
Exception Handling & File Handling	50	50	100
Modules, Packages, Iterators, Generators, Decorators	50	50	100
Multithreading, Multiprocessing, GIL, Memory Management	50	50	100

Total

- **Theory Questions: 350**
- **Coding Questions: 350**
- **Grand Total: 700 Interview Questions**

This would essentially be a **complete Python interview preparation guide** from beginner to advanced.

I suggest making it even better

Instead of only listing questions, I'll create a complete interview handbook for each topic that includes:

Part 1: Theory Questions (Top 50)

- Most frequently asked interview questions
- Ordered from basic to advanced
- Includes HR/tricky interview questions

Part 2: Coding Questions (Top 50)

- Easy
- Medium
- Hard
- FAANG-style problems
- Pattern-based questions

Part 3: Important Built-in Functions

Example for Lists:

- `append()`
- `extend()`

- insert()
- pop()
- remove()
- sort()
- reverse()
- copy()
- clear()

Part 4: Most Important Interview Programs

For example:

- Reverse string
- Palindrome
- Remove duplicates
- Find second largest
- Anagram
- Frequency count
- Rotate array
- Matrix problems

Part 5: Tricky Interview Questions

Examples:

- Why are tuples immutable?
- Why are dictionaries implemented as hash tables?
- Why is Python slower than C?
- Explain shallow vs deep copy.

Part 6: Frequently Asked Company Questions

Questions commonly asked in interviews at companies such as:

- TCS
- Infosys
- Wipro
- Cognizant
- Capgemini
- Accenture

- IBM
- Deloitte
- Oracle
- Amazon
- Microsoft
- Google

Part 7: Expected Interview Answers

For every theory question, I'll provide:

- Short answer (1–2 minutes)
- Detailed answer
- Example
- Common follow-up questions

Part 8: Difficulty Level

Each question will be labeled:

-  Easy
-  Medium
-  Hard

Part 9: Time to Solve

Coding questions will include expected interview solving times:

- 5 min
- 10 min
- 20 min
- 30 min

Final Structure

Volume 1

Python Basics

- 50 Theory
- 50 Coding

Volume 2

Strings, Lists, Tuples, Sets, Dictionaries

- 50 Theory
- 50 Coding

Volume 3

Functions & Recursion

- 50 Theory
- 50 Coding

Volume 4

OOP

- 50 Theory
- 50 Coding

Volume 5

Exception Handling & File Handling

- 50 Theory
- 50 Coding

Volume 6

Modules, Packages, Iterators, Generators, Decorators

- 50 Theory
- 50 Coding

Volume 7

Multithreading, Multiprocessing, GIL & Memory Management

- 50 Theory
- 50 Coding

Result

You'll receive a **700-question Python Interview Master Handbook**, organized from basic to advanced, suitable for:

- Campus placements
- Internship interviews
- Software Developer interviews
- Python Developer interviews
- Backend Developer interviews

- AI/ML interviews (Python fundamentals)
- Product-based company interviews

This handbook will be comprehensive enough that, after mastering it, you'll be well prepared for most Python interview rounds.